SHRI KRISHNASWAMY COLLEGE FOR WOMEN

ANNA NAGAR, CHENNAI - 600 040

(Affiliated to University of Madras)

Accredited by NAAC with B+ Grade 2(f) Status under UGC Act

ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APPLICATION

A Mini Project Submitted in partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF COMPUTER SCIENCE

BY

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MARCH 2025

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DECLARATION

I hereby declare that the project work entitled **"ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APPLICATION"** is a record of an original work done by me under the guidance of **Mrs. K.SHANTHI MCA., M.Phil., NET&SET,(Ph.D), ASSISTANT PROFESSOR,** PG Department of Computer Science, **SHRI KRISHNASWAMY COLLEGE FOR WOMEN** and this project work is submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Science. The results embodied in this project have not been submitted to any other university or Institute for the award of any degree or diploma.

DATE:

PLACE: **SIGNATURE OF THE STUDENT**

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ACKNOWLEDGEMENT

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ORGANIZATION PROFILE

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Triple M Infotech Solutions Pvt Ltd is a dynamic player in the tech industry, based in Chennai, India. Incorporated on September 7, 2010, it operates as a private limited company with a focus on web development, design, mobile app development, and search engine optimization. With an authorized share capital of INR 10.00 lac and a paid-up capital of INR 3.01 lac, Triple M Infotech stands as a testament to robust business growth and potential.

The company prides itself on offering a comprehensive suite of services that cater to the digital needs of its clients. From crafting static and dynamic websites to developing responsive e-commerce solutions, TripleM Infotech ensures that businesses make a significant online impact. Their expertise extends to digital marketing, where they assist startups in branding and online presence, ensuring that companies register effectively with government bodies.

Triple M Infotech also specializes in hosting services, managing web applications, and emails on their cloud server, providing a range of packages to suit various business needs. Their commitment to innovation is evident in their offerings of SEO, SEM, SMM, and on-demand online training through their e-learning portal.

With a vision to empower businesses through technology, Triple M Infotech has successfully completed numerous projects, leaving a trail of satisfied clients and a growing portfolio that speaks volumes of their dedication and expertise in the field.

ABSTRACT

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This project introduces an Online Medical Consultation and Subscription App, designed primarily as an appointment booking website that facilitates doctor-patient communication through text messaging and direct phone calls. Developed using HTML, CSS, and JavaScript, with MongoDB for admin management and database, the platform provides a streamlined experience for users to schedule appointments and subscribe to consultation plans for continuous healthcare support. The system ensures a seamless and intuitive user interface, allowing patients to browse doctor profiles, book appointments, and receive automated notifications. The integration of MongoDB enables efficient storage and management of user data, ensuring privacy and security. By removing the need for video consultations, this platform offers a cost-effective and accessible alternative for individuals seeking prompt medical advice. The project aims to bridge the gap between patients and healthcare providers by delivering a reliable, scalable, and user-friendly solution for modern medical consultations. Security and privacy are key considerations, with strong measures in place to protect patient data and ensure confidential communication between doctors and patients. By offering a cost-effective and accessible alternative to traditional telemedicine, this platform provides an innovative approach to digital healthcare. The project aims to bridge the gap between healthcare providers and patients, making medical consultations more convenient, efficient, and widely available.

INTRODUCTION

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In today’s fast-paced world, access to healthcare services has become a significant challenge due to time constraints, geographical barriers, and the increasing demand for medical consultations. The rapid advancement of digital technology has paved the way for online medical consultation platforms, offering patients a convenient and efficient way to connect with healthcare professionals. This project introduces an Online Medical Consultation and Subscription App, designed as an appointment booking website that enables doctor-patient communication through text messaging and direct phone calls.

The primary goal of this platform is to simplify the process of booking medical appointments while ensuring accessibility for individuals seeking quick and effective healthcare solutions. Unlike traditional telemedicine applications that rely on video conferencing, this system provides a direct and straightforward approach by allowing users to interact with doctors via text or phone calls. This feature makes the platform more inclusive and adaptable, particularly for individuals who may not have access to high-speed internet or are uncomfortable with video consultations.

Developed using HTML, CSS, and JavaScript, with MongoDB for admin management and database, the platform ensures a seamless user experience for both patients and administrators. The system allows users to browse doctor profiles, schedule appointments based on availability, and subscribe to medical consultation plans for ongoing healthcare support. The backend, powered by MongoDB, efficiently manages user data, doctor information, and appointment records, ensuring smooth and secure operations.

The platform prioritizes data security and privacy, ensuring that all medical interactions remain confidential. Automated notifications and reminders help patients keep track of their appointments, reducing missed consultations and enhancing overall user engagement. By eliminating the need for video calls, the system provides a cost-effective and easily accessible solution for patients while allowing healthcare providers to expand their reach.

This project aims to bridge the gap between patients and medical professionals, offering a modern, scalable, and user-friendly alternative to traditional healthcare services. By leveraging web-based technologies and database management, the platform enhances accessibility, efficiency, and reliability in medical consultations, ultimately contributing to the improvement of digital healthcare services.

PROJECT DESCRIPTION

PROJECT DESCRIPTION

AIM:

This project aims to develop an Online Medical Consultation and Subscription App for appointment booking via text messages and phone calls. Built with HTML, CSS, JavaScript, and MongoDB, it ensures secure data management, automated notifications, and easy accessibility, offering a cost-effective and user-friendly healthcare solution.

OBJECTIVE:

The objective of this project is to develop an Online Medical Consultation and Subscription App that allows patients to book appointments via text messages and phone calls. Using HTML, CSS, and JavaScript for the front end and MongoDB for admin management, the platform ensures secure data handling, seamless appointment scheduling, and subscription-based consultations. It aims to enhance accessibility, automate notifications, and prioritize data privacy, providing a cost-effective, user-friendly, and scalable solution for modern healthcare services while ensuring efficient doctor-patient communication.

FEATURES:

Online Appointment Booking:

Patients can book medical consultations through an easy-to-use interface. The system records their details, preferred consultation times, and subscription status, ensuring seamless scheduling.

Text-Based Consultation:

Communication between patients and doctors is conducted through messages and phone calls. This eliminates the need for video calls or physical visits, making consultations more accessible.

Admin Dashboard:

A dedicated admin panel allows authorized personnel to manage appointments, subscriptions, and patient records. Admins can approve, reschedule, or cancel bookings as needed.

Subscription Management:

Patients can opt for subscription-based consultations, providing them with ongoing medical support. The system tracks subscription status, renewal dates, and payment records.

Mongo DB Database:

A NoSQL database stores patient information, appointment details, and subscription records efficiently. Its document-based structure ensures scalability and flexibility in data management.

Authentication and Security:

The system includes login authentication for admins to ensure secure access. Only authorized users can modify records, preventing unauthorized changes or data breaches.

Scalability and Performance:

Designed to handle multiple users simultaneously, the system ensures smooth performance even as the number of appointments and subscriptions increases.

Responsive Design:

Developed with HTML, CSS, and JavaScript, the platform ensures compatibility across various devices, including desktops, tablets, and mobile phones.

Easy Backend Integration:

Node.js is used to handle backend operations, processing requests efficiently. MongoDB ensures smooth database interaction, allowing real-time updates of patient records and appointments.

Efficient Data Retrieval:

Fast and optimized database queries enable quick access to patient and appointment details. This reduces waiting times and improves overall system efficiency.

LITERATURE REVIEW

LITERATURE REVIEW

1.A Systematic Review of Online Medical Consultation Research

Publication Date: Approximately 7 months ago

Journal: Healthcare

Volume: 12, Issue 17, Article 1687

This systematic review examines the landscape of online medical consultation services, focusing on their utility and value in facilitating interactions between patients and doctors. The study retrieved and analysed 4,072 English records, ultimately including 75 articles. The research themes identified include patients utilizing online medical consultation platforms, doctors participating in these platforms, patients' choice of doctors, services provided by doctors, online patient reviews, service quality for patients, rewards to doctors, and the spill over effect between online and offline medical services. These themes form a theoretical framework encompassing the initiation, process, and outcomes of online medical consultations, providing a comprehensive understanding of the field and laying a foundation for future research.

2. Investigating Patient Use and Experience of Online Appointment Booking in General Practice

Publication Date: Approximately 8 months ago

Journal: Journal of Medical Internet Research

Volume: 26, Article e51931

This mixed-methods study explores how patients in England use and experience online appointment booking in general practice. Analysing data from the General Practice Patient Survey and conducting semi structured interviews, the study finds that engagement with online booking is influenced by factors such as the patient's practice, presence of long-term conditions, and socioeconomic status. The research highlights the need to consider these factors when designing and implementing online appointment systems to ensure equitable access and satisfaction among diverse patient groups.

3. Stake holders' Experiences, Perceptions and Satisfaction with an Electronic Appointment System: A Qualitative Content Analysis

Author: Faezeh Ostadmohammadi

Publication Date: Approximately 1 month ago

Journal: BMC Health Services Research

Volume: 25, Article 220

This qualitative study investigates the experiences, perceptions, and satisfaction of stakeholders with an electronic appointment system. Through interviews with patients and healthcare providers, the research identifies challenges and benefits associated with the system, offering insights into areas for improvement and strategies to enhance user satisfaction and system efficiency.

4. Online Medical Consultation Service–Oriented Recommendations

Publication Date: Approximately 8 months ago

Journal: Journal of Medical Internet Research

Volume: 26, Issue 1, Article e46073

This review analysed studies related to healthcare-related recommendations for online services, aiming to gather comprehensive evidence for evaluating current studies, identifying successful paradigms and approaches in service-oriented recommendations, and informing public health interventions and policy making. The study emphasizes leveraging two-sided recommendation technologies to enhance the well-being of both patients and physicians in the emerging online medical consultation service industry.

5. Evaluation of Patient Satisfaction With the New Web-Based Medical Appointment System

'Mawid' Program

Publication Date: Approximately 8 months ago

Journal: Cureus

Volume: 14, Issue 12, Article e32816

This study assessed patients' satisfaction and perception of the new web-based medical appointment system "Mawid" program at Primary Health Care Center in Jazan, Southwest Saudi Arabia. The findings revealed a high level of satisfaction among participants, with 94.3% expressing overall satisfaction. The system effectively improved patients' satisfaction with registration and reduced waiting times. The study suggests that patient satisfaction can be regularly assessed and used systematically as a quality and benchmarking instrument in primary health care.

6. Web-Based Medical Appointment Systems: A Systematic Review

Publication Date: Approximately 7.9 years ago

Journal: Journal of Medical Internet Research

Volume: 19, Issue 4, Article e134

This systematic review evaluates the benefits and barriers associated with implementing web-based medical appointment systems. Analyzing 36 articles discussing 21 different systems, the study found that most practices experienced positive outcomes, such as reduced no-show rates, decreased staff labor, shorter waiting times, and improved patient satisfaction. However, challenges like concerns over cost, flexibility, safety, and data integrity were identified as significant deterrents for providers considering the adoption of web-based scheduling. Additionally, patients' reluctance was often influenced by their prior experiences with technology and their communication preferences. The review highlights a growing trend toward adopting web-based appointment systems and underscores the need for further research to fully understand their impact on patient outcomes and healthcare efficiency.

SYSTEM ANALYSIS

SYSTEM ANALYSIS

EXISTING SYSTEM:

The existing system for medical consultations and appointment booking primarily relies on traditional methods such as in-person visits, phone call bookings, and limited online scheduling through hospital or clinic websites. Many healthcare facilities still use manual appointment scheduling, where patients must call or visit a clinic to book an appointment, leading to long waiting times, inefficiencies, and limited accessibility. Some hospitals and private healthcare providers offer basic online booking systems, but these often lack features such as real-time doctor availability, automated reminders, or secure patient-doctor communication. Additionally, existing platforms that support online consultations often require video calls, which may not be accessible to all patients due to internet limitations, technical issues, or privacy concerns. The lack of integrated, user-friendly, and efficient appointment and consultation systems creates difficulties in patient management, data security, and healthcare accessibility, highlighting the need for a more advanced and streamlined solution.

PROPOSED SYSTEM:

The proposed system is an Online Medical Consultation and Subscription App that allows patients to book appointments through text messages and phone calls, eliminating the need for in-person visits or complex online registration. Developed using HTML, CSS, and JavaScript for the front end and MongoDB for database management, the system ensures efficient appointment scheduling, secure data storage, and easy admin management. Unlike existing systems, it provides a simplified and accessible booking process without requiring video calls, making it suitable for users with limited internet access. The system also includes automated appointment reminders, subscription-based consultation options, and encrypted patient-doctor communication for enhanced privacy and security. By improving accessibility, streamlining appointment management, and reducing operational inefficiencies, the proposed system aims to provide a user-friendly, cost-effective, and scalable solution for modern healthcare services.

THE ROLE OF ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APPLICATION :

Telemedicine has revolutionized healthcare by providing remote medical consultations, reducing the need for in-person visits, and increasing accessibility for patients worldwide. With advancements in digital communication and cloud-based databases, online consultation platforms allow patients to connect with doctors through messaging and phone calls, ensuring timely medical advice. This approach is particularly beneficial for those in rural areas or with mobility constraints, offering them convenient healthcare solutions. Additionally, subscription-based models enhance patient engagement by providing ongoing medical support, making healthcare services more structured and affordable. As technology continues to evolve, telemedicine will play an increasingly vital role in improving efficiency, affordability, and accessibility in healthcare. By integrating secure databases like MongoDB and real-time appointment booking systems, telemedicine platforms ensure efficient management of patient records and consultations. As more healthcare providers adopt digital solutions, the future of telemedicine promises greater innovations, enhancing the overall patient experience and streamlining medical services.

USER REQUIREMENT ANALYSIS IN ONLINE MEDICAL CONSULTATION SYSTEMS:

System analysis plays a crucial role in developing efficient and user-friendly online medical consultation platforms by understanding user requirements and system functionality. A thorough user requirement analysis helps identify key features such as appointment booking, secure messaging, subscription management, and database integration to enhance the overall user experience. By gathering feedback from patients, doctors, and administrators, developers can ensure the system meets user expectations, improves accessibility, and streamlines healthcare services. Additionally, analysing security, scalability, and performance requirements helps in designing a robust system that handles increasing user traffic while ensuring data privacy and compliance with medical regulations.

DESIGN METHODOLGY:

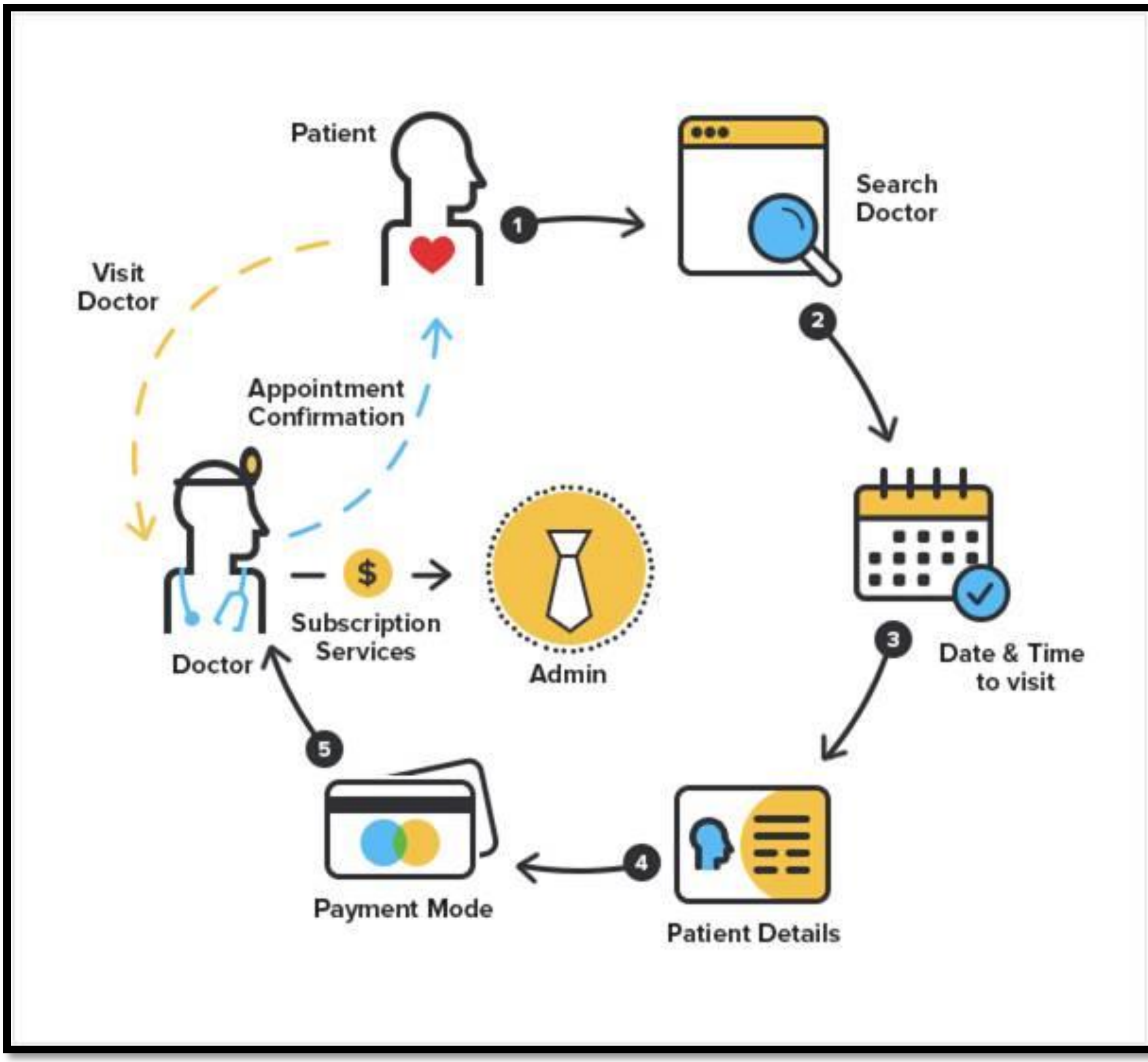
The Online Medical Consultation and Subscription App is developed as a web-based system using HTML, CSS, and JavaScript for structuring and managing the front-end, while MongoDB serves as the backend for admin management and database storage. Unlike traditional web-based appointment booking systems, this application does not provide a direct UI for patients; instead, all communication occurs via text messages and phone calls. Patients can book appointments by sending a message or calling the provided contact number, and their requests are stored and managed in the MongoDB database by the admin.

The backend logic is responsible for processing appointment requests, handling subscription-based consultations, and updating the appointment status in real-time. The admin dashboard is built to allow easy appointment tracking, patient management, and consultation scheduling, ensuring that all interactions are recorded securely. Authentication and role-based access control are implemented to restrict access to sensitive medical data, ensuring that only authorized admins can modify or view patient records.

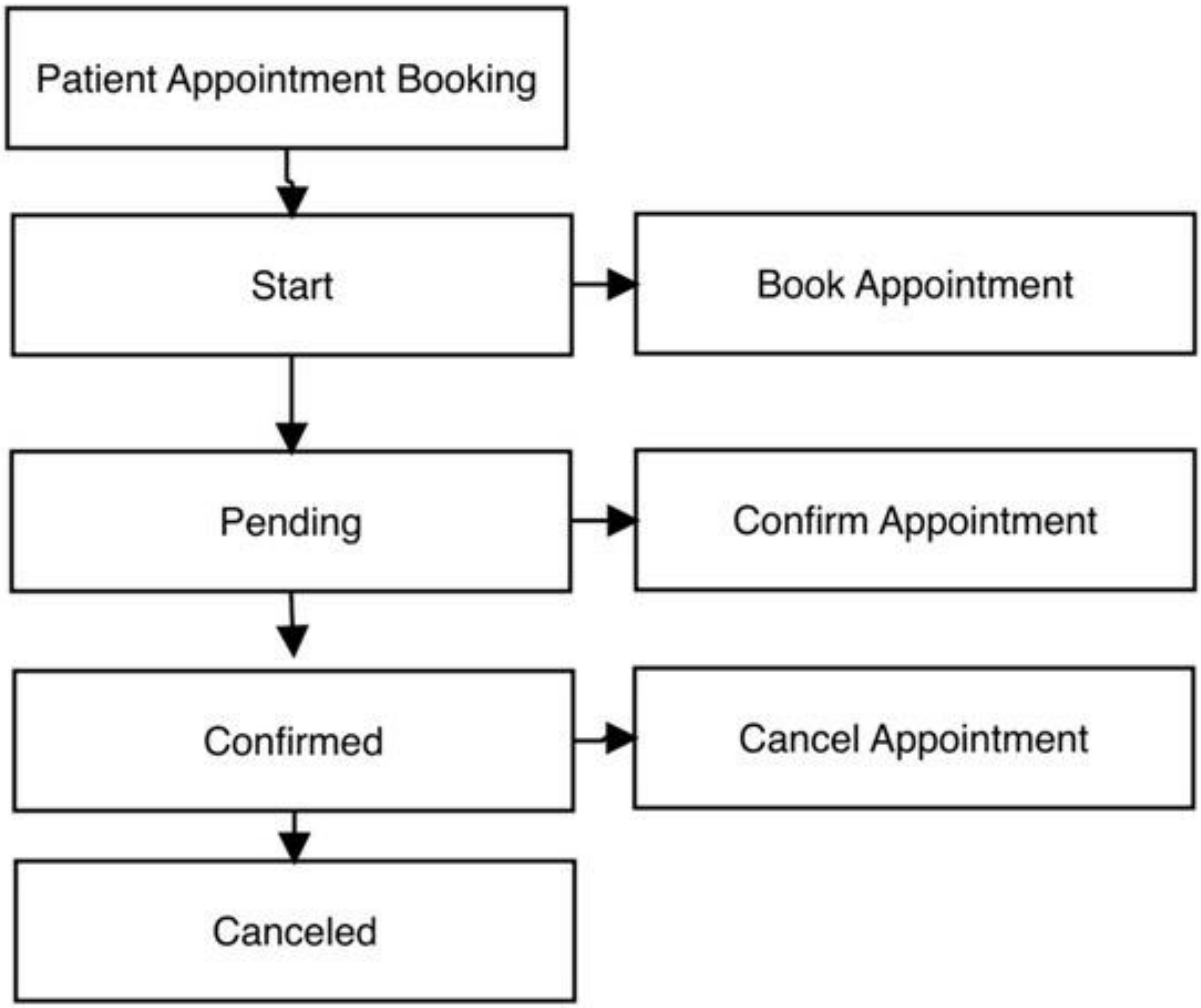
The system follows a modular architecture, where different components handle specific functionalities, such as appointment booking, patient record management, and notifications. Automated SMS-based confirmations and reminders are integrated to keep both patients and doctors updated about upcoming consultations. The use of MongoDB ensures efficient data retrieval and management, allowing quick access to patient details and appointment history.

The development process follows an incremental approach, allowing improvements and modifications based on system performance and user feedback. The system is optimized for scalability, ensuring that more doctors and patients can be added as needed. Regular testing and debugging are performed to maintain system reliability, ensuring that appointment scheduling and consultation management operate smoothly without UI dependency.

SYSTEM ARCHITECTURE


Patient
Search Doctor
Visit Doctor
Appointment Confirmation
Date & Time to visit
Doctor
Subscription Services
Admin
Payment Mode
Patient Details
1
2
3
4
5

FLOW CHART:



Patient Appointment Booking
Start
Book Appointment
Pending
Confirm Appointment
Confirmed
Cancel Appointment
Canceled

ADVANTAGES:

- Eliminates the need for a graphical user interface (UI), allowing patients to book appointments via text messages and phone calls.
- Provides a simple and accessible booking process without requiring technical knowledge or internet access.
- Ensures fast and reliable communication through direct phone-based consultations.
- Uses a Mongo DB powered database for efficient admin management of appointments, patient records, and subscriptions.
- Sends automated SMS notifications for appointment confirmations and reminders, reducing missed consultations.
- Features a modular architecture, making it scalable for adding more doctors and patients as needed.
- Enhances data security with authentication and access controls for authorized admin management.
- Reduces operational costs by eliminating the need for complex UI development and video-based consultations.
- Improves accessibility for users in rural areas or those with limited internet access, requiring only a mobile phone for interaction.
- Saves time for both doctors and patients by minimizing waiting times and reducing administrative workload.

SYSTEM REQUIREMENT SPECIFICATION

SYSTEM REQUIREMENT SPECIFICATION

1. HARWARE REQUIREMENTS:

The hardware requirements may serve as the basic for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by the software engineers as the string point of the system design.

PROCESSOR	12th Gen Intel (R) Core(TM) i3-1215U 1.20 GHz
RAM	8 GB
SYSTEM TYPE	64-Bit operating system, x64-based processor
STORAGE	Solid-State Drive (SSD)

2. SOFTWARE REQUIREMENTS:

The software requirements are the specification of the system. It should include both the definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the team's progress throughout the development activity.

OPERATING SYSTEM	Windows 11	
SOFTWARE	Front-end Development	▪ HTML ▪ CSS
	Bank-end Development	JAVASCRIPT
	Data Base	MONGODB

SOFTWARE DESCRIPTION

SOFTWARE DESCRIPTION

Introduction to HTML:

HTML (HyperText Mark-up Language) is the standard language used to create and structure web pages. It defines the basic building blocks of a webpage using elements and tags, which organize content such as text, images, links, and multimedia. Developed by Tim Berners-Lee and first introduced in 1993, HTML has evolved through multiple versions, with HTML5 being the latest, offering enhanced features like semantic elements, multimedia support, and improved accessibility. HTML works alongside CSS (Cascading Style Sheets) for styling and JavaScript for interactivity, forming the foundation of modern web development. As a platform-independent language, HTML allows content to be displayed consistently across different browsers and devices, making it an essential tool for creating responsive and user-friendly websites.

Introduction to CSS:

CSS (Cascading Style Sheets) is a style sheet language used to control the presentation, layout, and design of web pages. It works alongside HTML to define how elements should be displayed, including aspects like colours, fonts, spacing, positioning, and responsiveness. Introduced by the World Wide Web Consortium (W3C) in 1996, CSS has evolved through multiple versions, with CSS3 introducing features such as animations, gradients, flex box, and grid layouts, enabling modern and dynamic web design. CSS follows a cascade and inheritance model, allowing styles to be applied efficiently across multiple elements and pages. By separating content from design, CSS enhances maintainability, reusability, and accessibility, making web development more efficient and visually appealing.

Introduction to JAVASCRIPT:

JavaScript is a powerful scripting language widely used for developing dynamic and interactive web applications. In this project, JavaScript plays a crucial role in enhancing the user experience by enabling real-time interactions, handling form validations, and managing asynchronous data exchanges between the client and the server. As a client-side language, JavaScript ensures smooth appointment booking, user authentication, and seamless communication between patients and doctors through an intuitive interface. With JavaScript, key functionalities such as fetching and displaying appointment data, handling user inputs, and managing UI updates without page reloads are efficiently implemented. Combined with HTML and CSS, JavaScript creates a responsive and user-friendly experience, making it easier for patients to interact with the system.

Introduction to MONGODB:

Mongo DB is a No SQL, document-oriented database designed for high performance, scalability, and flexibility. Unlike traditional relational databases, it stores data in JSON-like BSON (Binary JSON) documents, allowing for dynamic and schema-less data storage. Developed by Mongo DB Inc. and first released in 2009, Mongo DB is widely used for web applications, big data, and real-time analytics due to its ability to handle large amounts of unstructured data efficiently. It supports collections instead of tables, where each document can have different fields, making it highly adaptable for modern applications. Mongo DB offers features like indexing, replication, sharing, and built-in security, ensuring fast data retrieval, redundancy, and scalability. As a cross-platform and cloud-compatible database, Mongo DB is widely used in web development, including applications that require real-time data processing, such as appointment booking systems, e-commerce platforms, and healthcare management applications

USES OF MONGODB:

- **Storing Patient and Appointment Data:**

  Mongo DB is used to store patient details, including name, contact numbers, and appointment date/time. It also manages appointment records, allowing the admin to access, update, or cancel bookings as needed.

- **Managing Subscription Plans:**

  The system allows patients to opt for subscription-based consultations, and Mongo DB efficiently stores subscription details, including plan type, validity, and payment status.

- **Real-Time Data Retrieval:**

  As a No SQL database, Mongo DB enables fast data retrieval for appointment scheduling, ensuring that the admin can quickly access patient records and appointment history without delays.

- **Admin Authentication and Access Control:**

  Mongo DB stores admin login credentials securely, ensuring that only authorized personnel can access and manage appointment records. Authentication mechanisms prevent unauthorized modifications.

- **Scalability for Future Expansion:**

  Mongo DB’s scalable architecture allows the system to grow as more patients and doctors are added. Its ability to handle increasing data volumes ensures smooth performance even with high user engagement.

- **Secure and Reliable Data Storage:**

  With features like replication and backup support, Mongo DB ensures that patient and appointment data is stored safely, minimizing the risk of data loss or corruption.

- **Integration with Node.js for Backend Operations:**

  Mongo DB seamlessly integrates with Node.js and Express.js, allowing efficient backend communication for handling appointment requests, subscription management, and patient records.

DOMAIN SPECIFICATION

DOMAIN SPECIFICATION

This project falls under the domain of **healthcare technology and telemedicine**, focusing on providing **remote medical consultation** and **appointment booking** through an efficient, secure, and user-friendly platform. The system allows patients to communicate with doctors via **text-based messaging and phone calls**, ensuring accessibility, convenience, and improved healthcare management. The project integrates **modern web technologies (HTML, CSS, JavaScript, Node.js, and MongoDB)** to offer a **scalable, secure, and high-performance solution** for online medical services.

KEY DOMAIN SPECIFICATION:

Healthcare Technology and Digital Transformation:

- Leverages digital platforms to provide online medical consultation services.
- Reduces dependency on physical visits, enhancing accessibility and convenience.

Telemedicine and Remote Consultation:

- Facilitates patient-doctor interaction through text-based messages and phone calls.
- Helps patients in remote or rural areas access medical assistance easily.
- Reduces waiting times and hospital overcrowding.

Appointment Scheduling and Management:

- Provides an online platform for booking, managing, and canceling medical appointments.
- Admin panel allows doctors or administrators to manage schedules efficiently.

Subscription-Based Healthcare Model:

- Patients can subscribe for long-term consultation services with flexible plans.
- Subscription tracking helps manage renewals and user engagement.

Database Management with MongoDB:

- Securely stores patient details, appointment records, and subscription data.
- Ensures efficient data retrieval and management.
- Uses NoSQL database architecture for scalability and high performance.

Scalability and Performance Optimization:

- Designed to support multiple users simultaneously without system slowdowns.
- Uses Node.js for backend operations, ensuring smooth request handling.
- MongoDB integration optimizes data processing and retrieval.

Security, Privacy, and Authentication:

- Implements secure login authentication for admin and authorized users.
- Ensures patient data privacy, complying with healthcare regulations.
- Prevents unauthorized access to sensitive medical information.

User-Friendly Web Interface

- Developed with HTML, CSS, and JavaScript for a responsive and interactive UI.
- Optimized for accessibility across different devices, including desktops, tablets, and mobile phones.

Communication and Notification System

- Patients receive notifications for appointment confirmations, reminders, and subscription renewals.
- Doctors can send important health updates or follow-up instructions to patients.

Cost-Effective and Time-Saving Healthcare Solution

- Reduces travel expenses and waiting time for patients.
- Helps doctors manage their consultations efficiently, improving overall service quality.

HOW DOES THE ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APPLICATION WORK?

User Registration and Login:

- Patients and administrators register on the platform with basic details.
- Secure login authentication ensures only authorized access.

Appointment Booking Process:

- Patients select an available doctor and book an appointment.
- The system records the booking details in the database.
- Users receive confirmation via the platform.

Text-Based and Phone Call Consultation:

- Patients communicate with doctors via text messages or phone calls.
- Doctors provide medical advice and prescriptions based on symptoms shared.

Subscription Management:

- Users can opt for subscription plans for long-term medical support.
- The system tracks subscription status and renewal dates.

Admin Panel for Management:

- Admins can view, approve, reschedule, or cancel appointments.
- Subscription records and patient details are managed efficiently.

MongoDB Database for Data Storage:

- Patient records, appointment details, and subscription data are stored securely.
- NoSQL structure ensures quick retrieval and updates.

Secure Authentication and Data Privacy:

- User credentials are encrypted for security.
- Access control ensures data protection and compliance with healthcare regulations.

Scalability and Performance Optimization:

- The system handles multiple users simultaneously.
- Node.js manages backend processes efficiently.

User Notifications and Updates:

- Patients receive appointment confirmations and reminders.
- Doctors can send follow-up instructions via the platform.

FUTURES OF ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APPLICATION:

AI-Powered Chatbot Assistance:

- Implementing an AI chatbot to provide instant responses to common medical queries before connecting to a doctor.

Video Consultation Integration:

- Adding video call functionality for more effective doctor-patient interaction beyond text and phone calls.

Electronic Prescription System:

- Enabling doctors to generate and send digital prescriptions directly through the platform.

Automated Appointment Reminders:

- Implementing SMS and email notifications to remind patients about upcoming consultations.

Multi-Doctor Consultation Feature

- Allowing patients to seek second opinions or consult multiple specialists for complex health issues.

Health Record Management:

- Introducing a feature for patients to upload and store their medical history, test reports, and prescriptions.

Payment Gateway for Online Transactions:

- Enabling secure online payment options for consultation fees and subscription plans.

Integration with Wearable Health Devices:

- Syncing data from fitness trackers and smartwatches to provide doctors with real-time health insights.

TECHNOLOGIES USED IN ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APPLICATION:

TECHNOLOGY	USAGE
HTML	Structure and content of the web pages.
CSS	Styling and layout of the user interface.
JAVASCRIPT	Creates interactive and dynamic web pages.

TYPES OF USERS & THEIR ACCESS LEVELS:

USERS	ACCESS PRIVILEGES
Administrator	Manage doctor, patients, Subscriptions and security details.
Patients	Register, book appointment, subscription details.
Doctors	Manage appointments, communicate with patients, provide medical advice, update consultation status.

APPLICATIONS OF ONLINE MEDICAL CONSULTATION AND SUBSCRIPTION APP:

Remote Healthcare Access:

- Enables patients to consult doctors from anywhere, reducing the need for physical visits.

24/7 Medical Assistance:

- Provides round-the-clock consultation services, ensuring timely medical advice.

Chronic Disease Management:

- Helps patients with long-term conditions schedule regular check-ups and receive continuous care.

Emergency Medical Guidance:

- Assists users with immediate medical concerns before visiting a hospital.

Subscription-Based Healthcare Services:

- Offers flexible **plans** for regular consultations and follow-ups.

Rural and Remote Area Healthcare:

- Improves access to medical services in underdeveloped regions with limited healthcare facilities.

Elderly and Disabled Patient Support:

- Allows those with mobility issues to receive medical care from home.

Doctor Consultation Convenience:

- Reduces waiting times and optimizes doctors' schedules with digital appointment management.

Secure Medical Records Management:

- Stores patient history and consultation details for easy access and follow-ups.

Cost-Effective Healthcare Solution:

- Lowers travel and consultation costs, making healthcare more affordable.

MODULE DESCRIPTION

MODULE DESCRIPTION

The app enables users to book appointments and consultation doctors via video, audio, or chat. Doctors manage profiles and issue digital prescriptions, while patients access medical records. It offers secure payments, automated reminders, and subscription options. Admins handle doctor verification, user management, and analytics. A support system ensures seamless user assistance.

- **User Management Module**
- **Doctor Management Module**
- **Appointment Booking Module**
- **Online Consultation Module**
- **Subscription & Payment Module**
- **Prescription & Medical Records Module**
- **Notification & Alerts Module**
- **Admin Panel Module**
- **Review & Feedback Module**
- **Help & Support Module**

1.User Management Module

Description: This module handles user registration, authentication, and profile management. It supports different user roles, including patients, doctors, and administrators.

Features:

- User signup/login with email and phone verification
- Role-based access control (patients, doctors, admin)
- Profile creation and editing
- Secure password management

2.Doctor Management Module

Description: This module allows doctors to register, verify credentials, and manage their availability.

Features:

- Doctor registration and approval process
- Profile setup (specialization, experience, consultation fees)
- Availability scheduling
- Review and ratings system

3. Appointment Booking Module

Description: Facilitates online booking and scheduling of medical consultations.

Features:

- Search and filter doctors by specialization
- Real-time appointment availability
- Booking confirmation and reminders
- Rescheduling and cancellation options

4. Online Consultation Module

Description: Enables virtual consultations via chat, voice, or video calls.

Features:

- Secure video/audio consultation platform
- Text-based chat support
- Real-time prescription generation
- Digital medical history storage

5. Subscription & Payment Module

Description: Manages subscription plans for premium consultations and other healthcare services.

Features:

- Multiple subscription plans (monthly, yearly)
- Secure online payment integration (credit/debit cards, UPI, wallets)

- Invoice generation and billing history
- Free trial or pay-per-use options

6. Prescription & Medical Records Module

Description: Allows doctors to generate and manage digital prescriptions and medical records.

Features:

- Electronic prescription generation
- Secure storage of patient medical history
- Downloadable and shareable prescriptions
- Integration with pharmacies for medicine ordering

7. Notification & Alerts Module

Description: Sends reminders, alerts, and important updates to users.

Features:

- Appointment reminders via email/SMS
- Subscription renewal alerts
- New consultation notifications
- Doctor availability updates

8. Admin Panel Module

Description: Provides an interface for administrators to manage users, doctors, and system settings.

Features:

- Dashboard for monitoring system activities
- Doctor approval and verification
- User account management
- Reports and analytics

9. Review & Feedback Module

Description: Allows patients to rate and review their consultation experiences.

Features:

- Doctor rating system
- Patient feedback collection
- Review moderation by admins

10. Help & Support Module

Description: Provides customer support and FAQs for users.

Features:

- In-app chat bot for instant assistance
- Live customer support chat
- FAQs and knowledge base

SYSTEM DESIGN

SYSTEM DESIGN

UML DIAGRAMS

The Unified Modeling Language (UML) is used to specify, visualize, modify, construct and document the artifacts of an object-oriented software intensive system under development. UML offers a standard way to visualize a system's architectural blueprints, including elements such as:

- actors
- business processes
- (logical) components
- activities
- programming language statements
- Reusable software components.

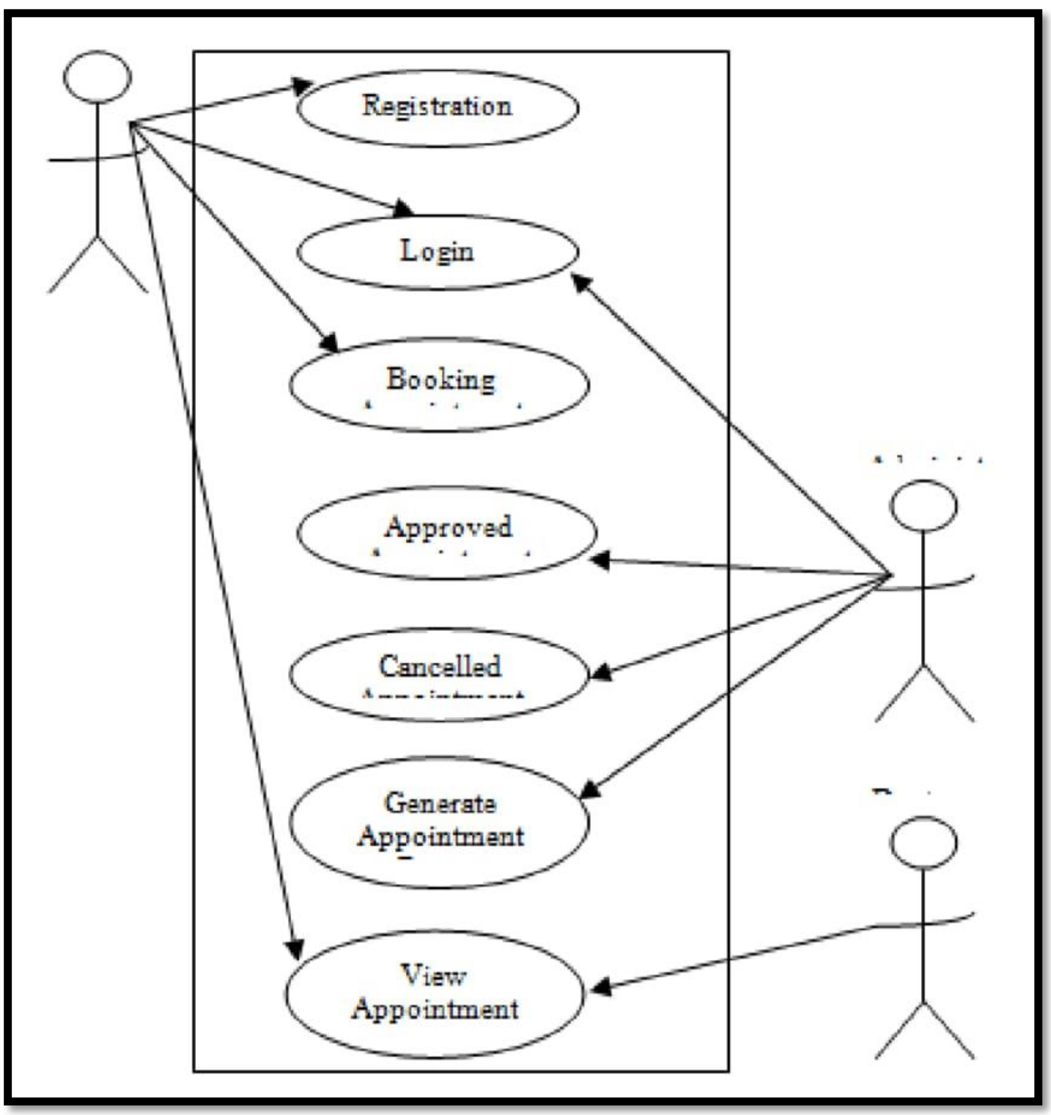
UML combines best techniques from data modeling (entity relationship diagrams), business modeling (work flows), object modeling, and component modeling. It can be used with all processes, throughout the software development life cycle, and across different implementation technologies. UML has synthesized the notations of the Booch method, the Object-modeling technique (OMT) and Object-oriented software engineering (OOSE) by fusing them into a single, common and widely usable modeling language. UML aims to be a standard modeling language which can model concurrent and distributed systems.

TYPES OF UML DIAGRAMS:

- Use case diagram
- Class diagram
- Activity diagram
- Sequence diagram

USE CASE DIAGRAM :

- UML is a standard language for specifying, visualizing, constructing, and documenting the artifacts of software systems.
- UML was created by Object Management Group (OMG) and UML 1.0 specification draft was proposed to the OMG in January 1997.
- OMG is continuously putting effort to make a truly industry standard.
- UML stands for **U**nified **M**odeling **L**anguage.
- UML is a pictorial language used to make software blue prints



CLASS DIAGRAM:

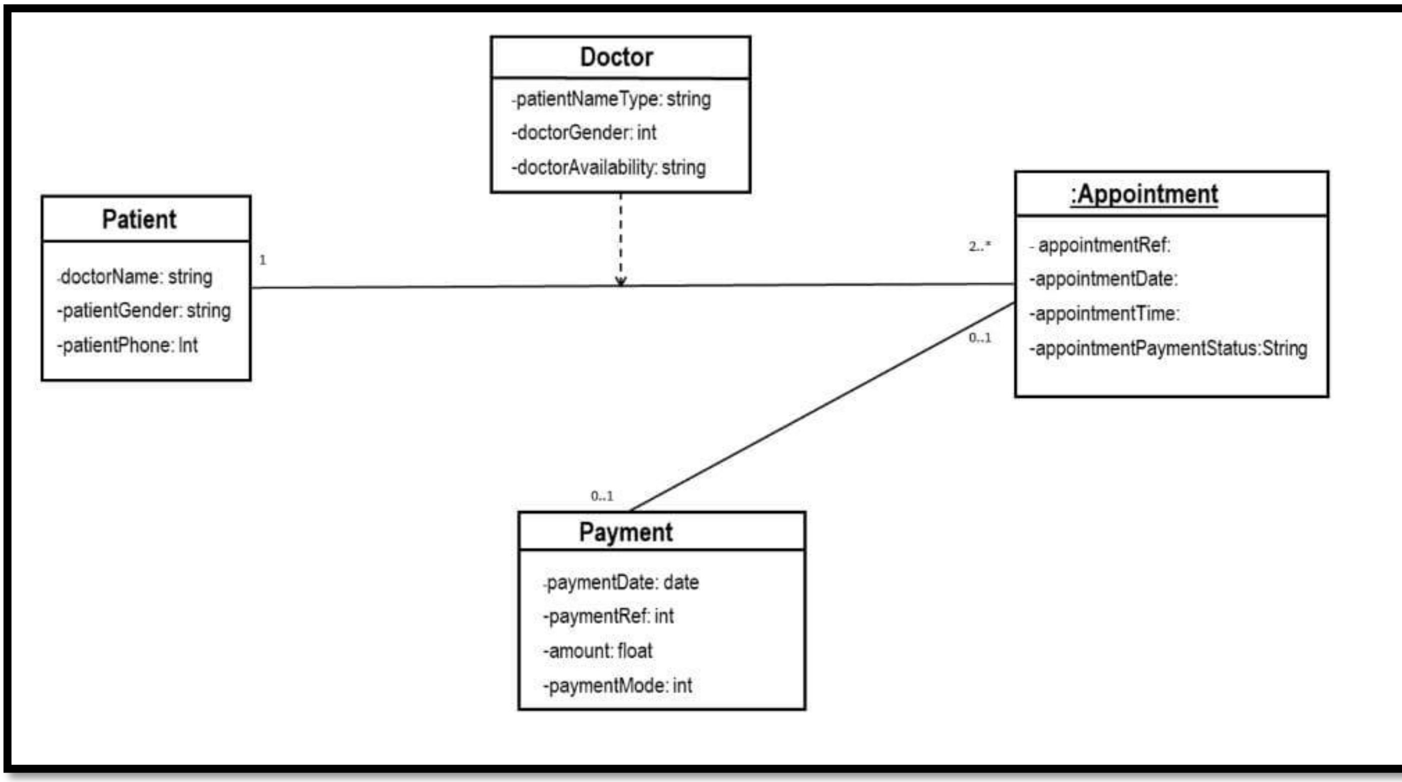
- The class diagram is the main building block of object-oriented modeling. It is used for general conceptual modeling of the systematic of the application, and for detailed modeling translating the models into programming code. Class diagrams can also be used for data modeling. The classes in a class diagram represent both the main elements, interactions in the application, and the classes to be programmed.

- In the diagram, classes are represented with boxes that contain three compartments:

    - The top compartment contains the name of the class. It is printed in bold and centered, and the first letter is capitalized.

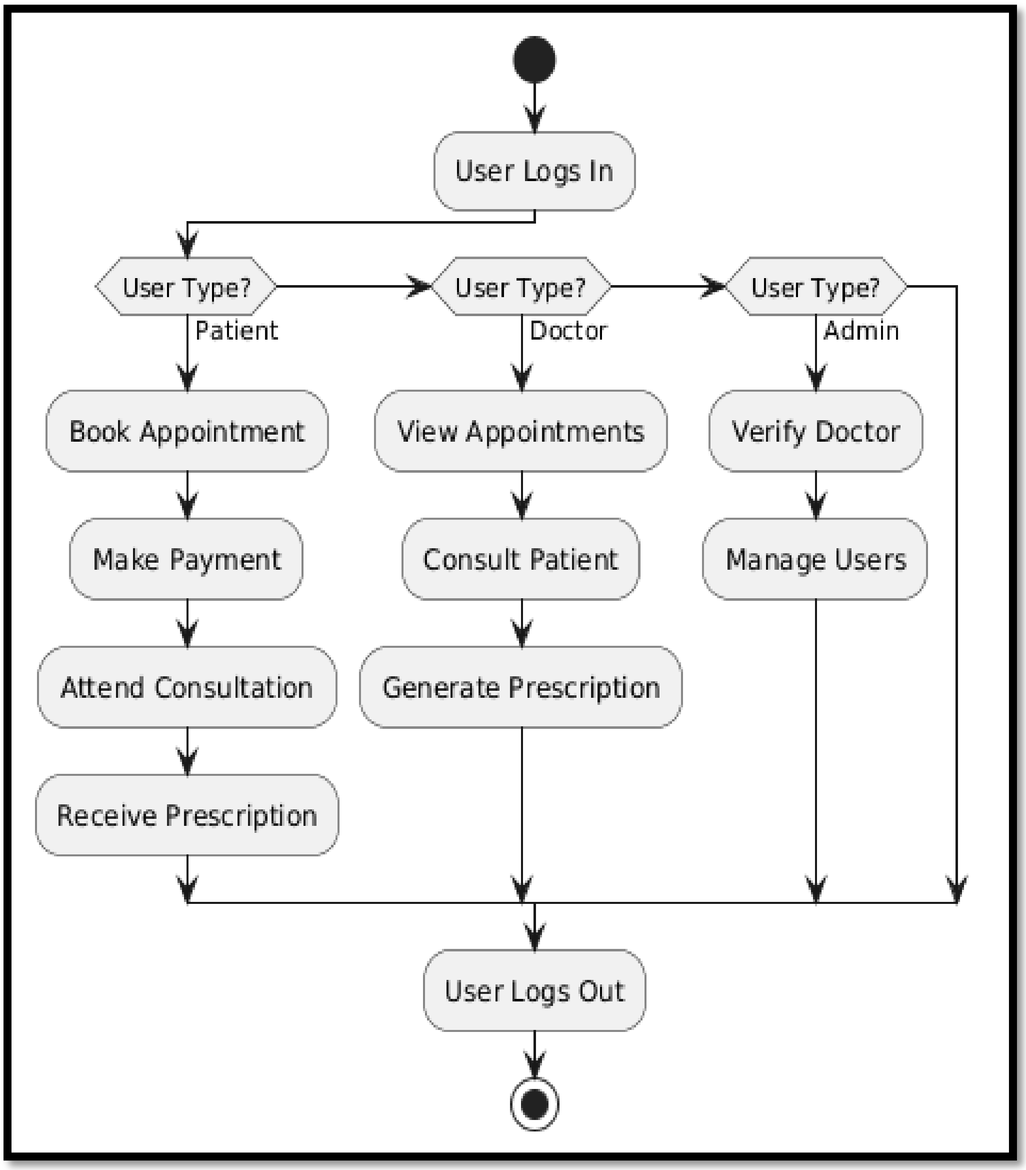
    - The middle compartment contains the attributes of the class. They are left-aligned and the first letter is lowercase.

    - The bottom compartment contains the operations the class can execute. They are also left-aligned and the first letter is lowercase.



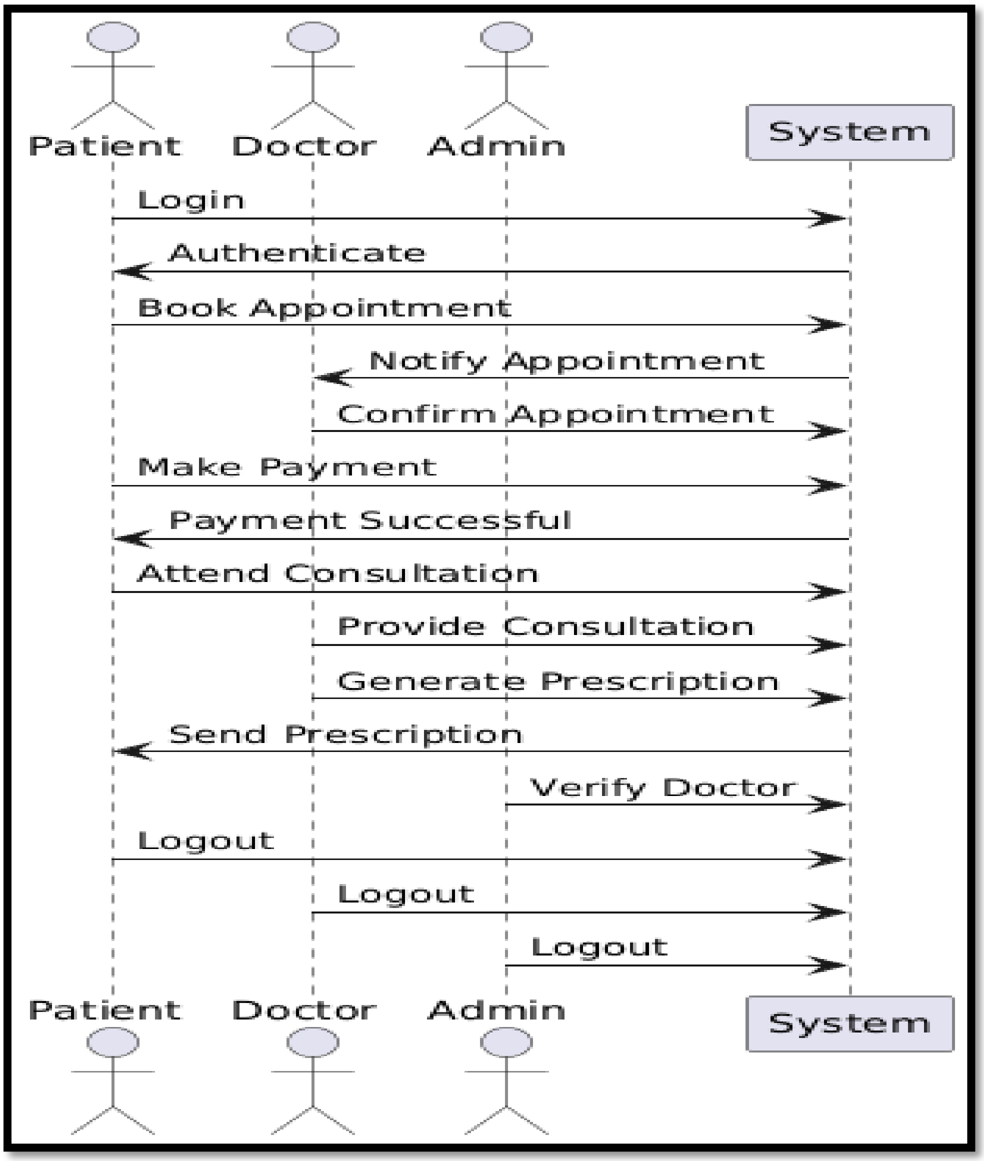
ACTIVITY DIAGRAM :

Activity diagrams are graphical representations of Workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.


User Logs In
User Type?
Patient
User Type?
Doctor
User Type?
Admin
Book Appointment
Make Payment
Attend Consultation
Receive Prescription
View Appointments
Consult Patient
Generate Prescription
Verify Doctor
Manage Users
User Logs Out

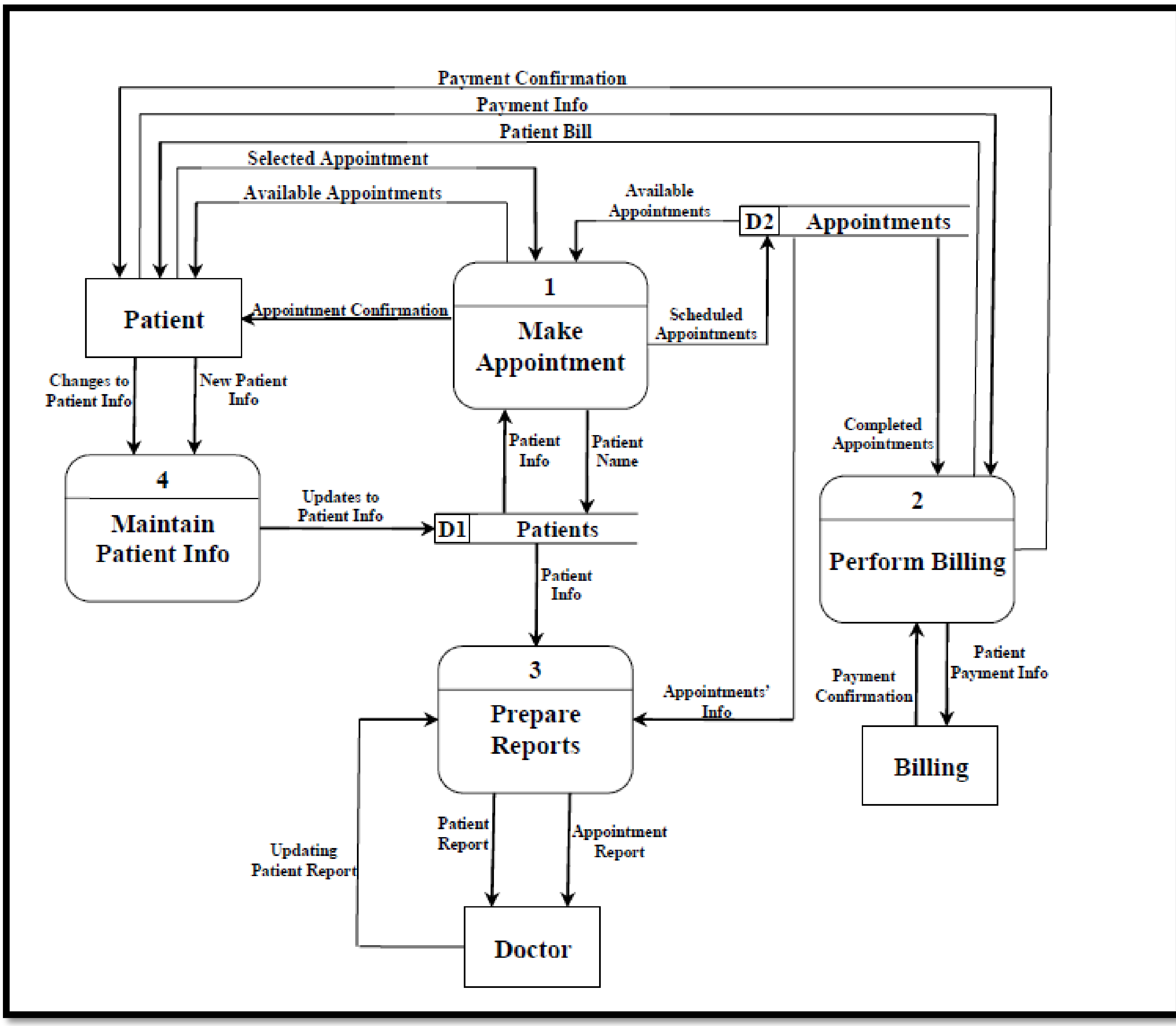
SEQUENCE DIAGRAM:

Sequence Diagrams Represent the objects participating the interaction horizontally and time vertically. A Use Case is a kind of behavioral classifier that represents a declaration of an offered behavior. Each use case specifies some behavior, possibly including variants that the subject can perform in collaboration with one or more actors. Use cases define the offered behavior of the subject without reference to its internal structure. These behaviors, involving interactions between the actor and the subject, may result in changes to the state of the subject and communications with its environment. A use case can include possible variations of its basic behavior, including exceptional behavior and error handling.

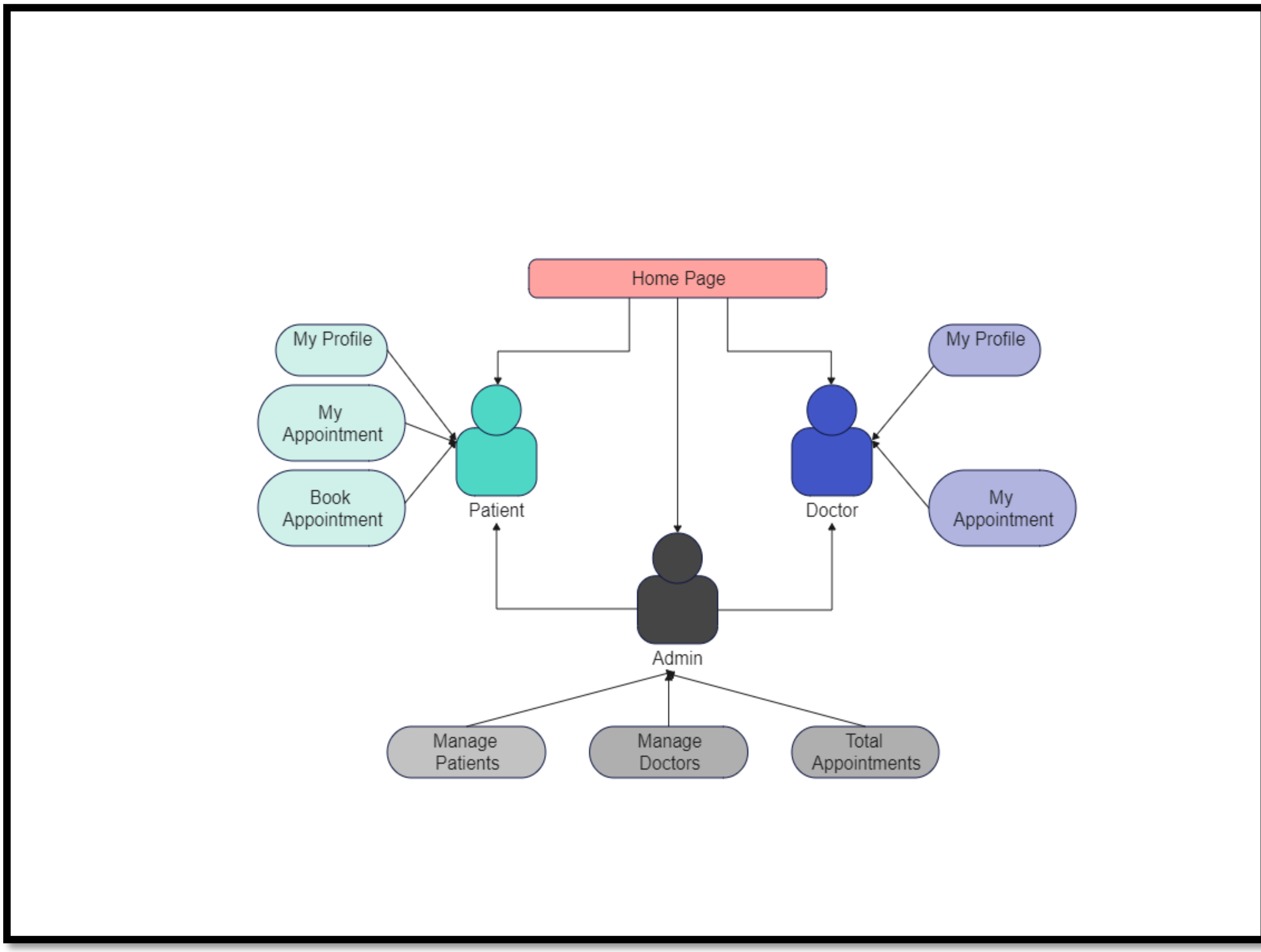


DATA FLOW DIAGRAMS:

The DFD takes an input-process-output view of a system i.e. data objects flow into the software, are transformed by processing elements, and resultant data objects flow out of the software.

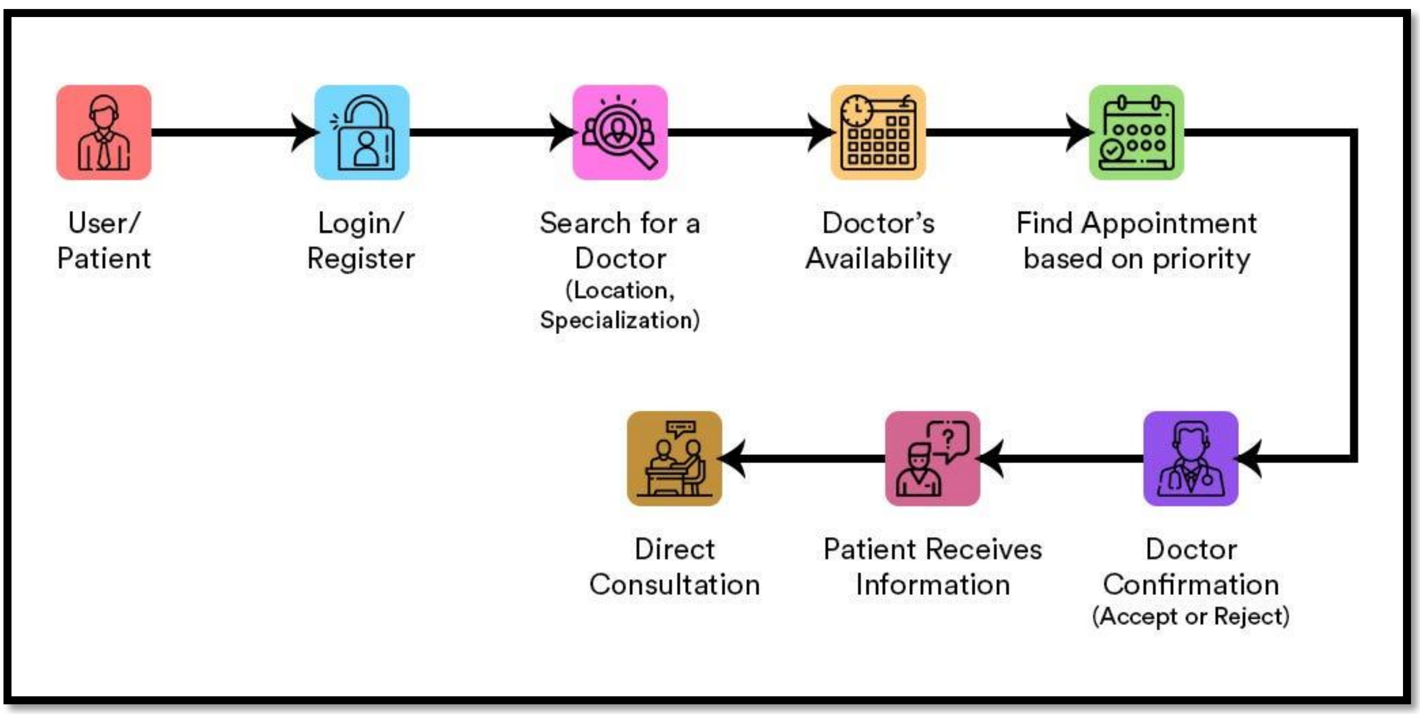


ROLE-BASED ACCESS DIAGRAM:



PATIENT CONSULTATION PROCESS FLOW:

This diagram represents the step-by-step process of how a patient interacts with an online medical consultation system. The flow begins with the patient registering or logging into the system, followed by searching for a doctor based on location and specialization. The system checks the doctor's availability, and an appointment is scheduled based on priority. The doctor receives the request and can accept or reject the appointment. Once confirmed, the patient receives consultation details and proceeds with the session. Finally, the patient receives information regarding the consultation, completing the process. This structured flow ensures a seamless, efficient, and accessible healthcare experience for both patients and doctors.



SOURCE CODE

SOURCE CODE

FRONTEND

HTML:

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/vite.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>ZeeCare</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

CSS:

```css
body {
    font-family: Arial, sans-serif;
    background-color: #f4f4f9;
    margin: 0;
    padding: 0;
}

.container {
    width: 80%;
    margin: auto;
    overflow: hidden;
}
```

```
h2 {
    color: #333;
}

form {
    margin-top: 20px;
}

input, button {
    display: block;
    width: 100%;
    padding: 10px;
    margin-bottom: 10px;
}

button {
    background-color: #5cb85c;
    color: white;
    border: none;
    cursor: pointer;
}

button:hover {
    background-color: #4cae4c;
}
```

BACKEND

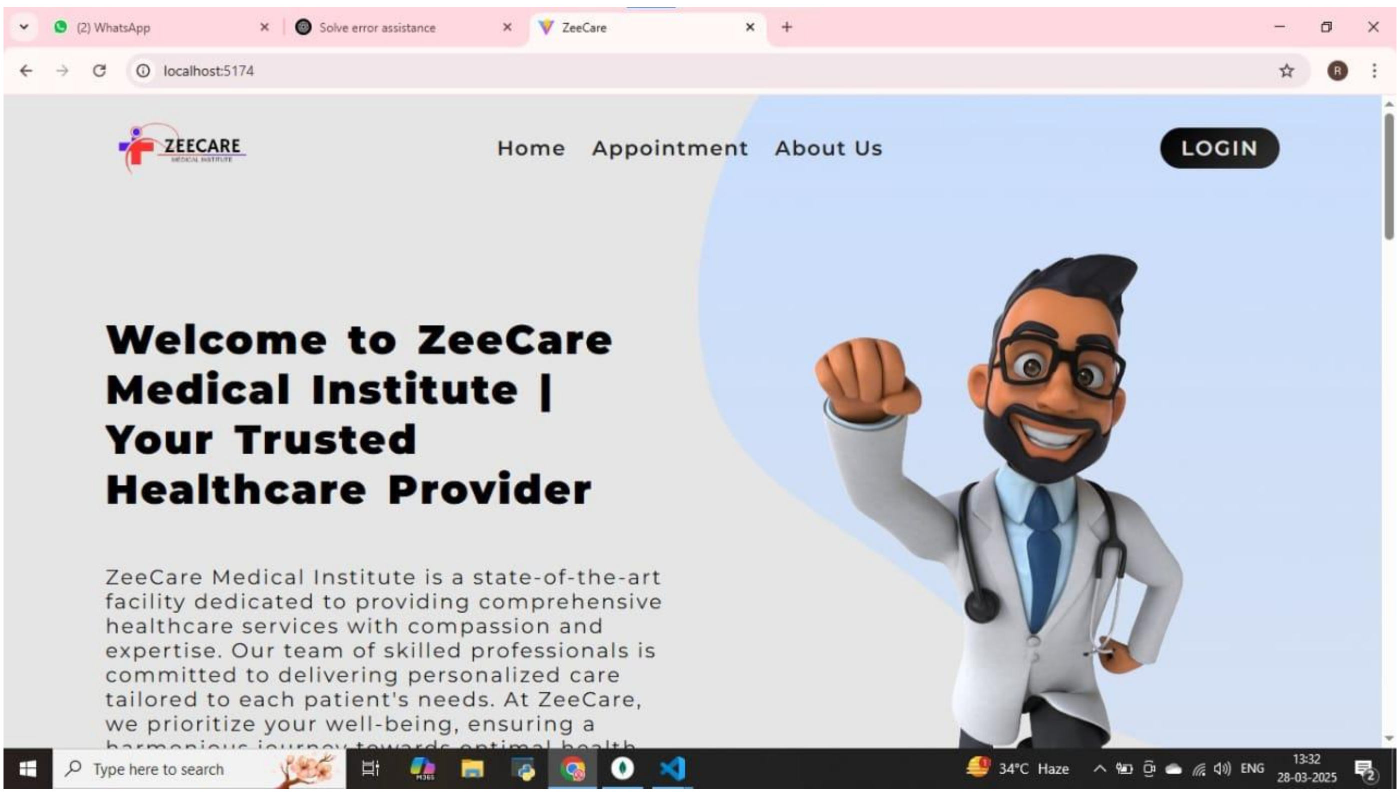
App.js:

```
import express from "express";
import { dbConnection } from "./database/dbConnection.js";
import { config } from "dotenv";
import cookieParser from "cookie-parser";
import cors from "cors";
import fileUpload from "express-fileupload";
import { errorMiddleware } from "./middlewares/error.js";
import messageRouter from "./router/messageRouter.js";
import userRouter from "./router/userRouter.js";
import appointmentRouter from "./router/appointmentRouter.js";

config();
const app = express();

app.use(cors());

app.use(cookieParser());
app.use(express.json());
app.use(express.urlencoded({ extended: true }));

app.use(
fileUpload({
useTempFiles: true,
tempFileDir: "/tmp/",
})
);
app.use("/api/v1/message", messageRouter);
```

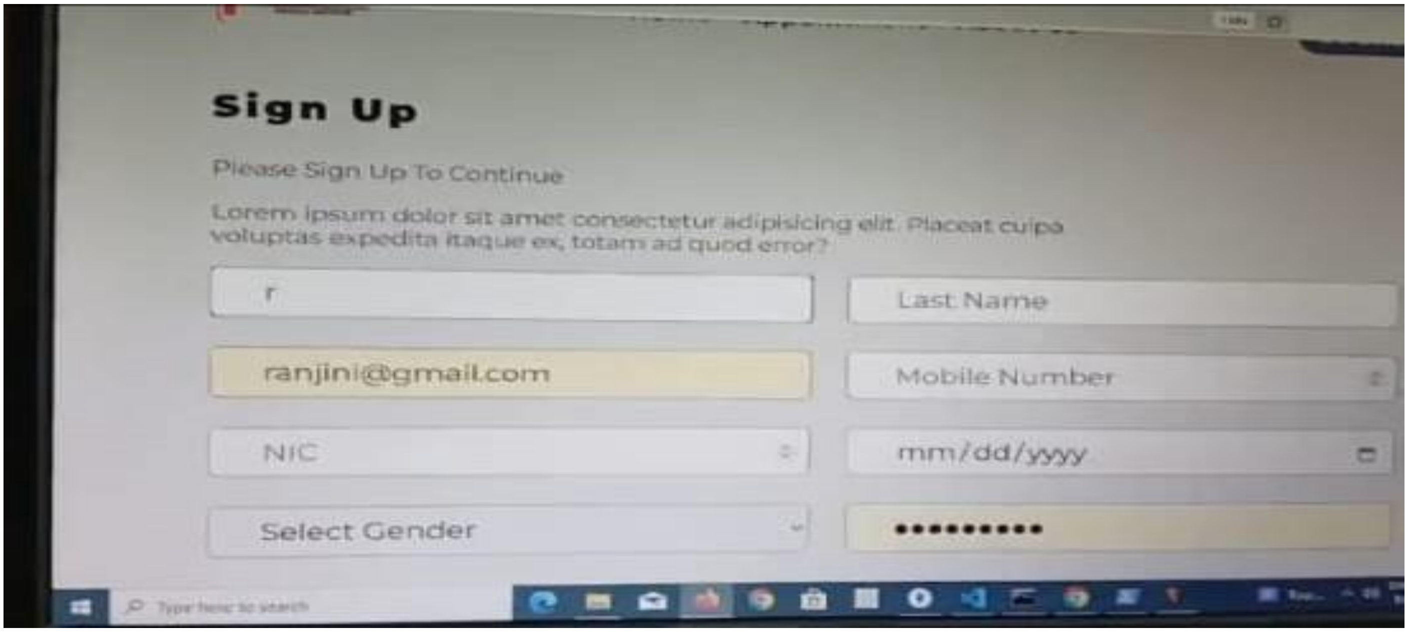
```
app.use("/api/v1/user", userRouter);
app.use("/api/v1/appointment", appointmentRouter);

dbConnection();

app.use(errorMiddleware);
export default app;
```

SCREEN SHOTS

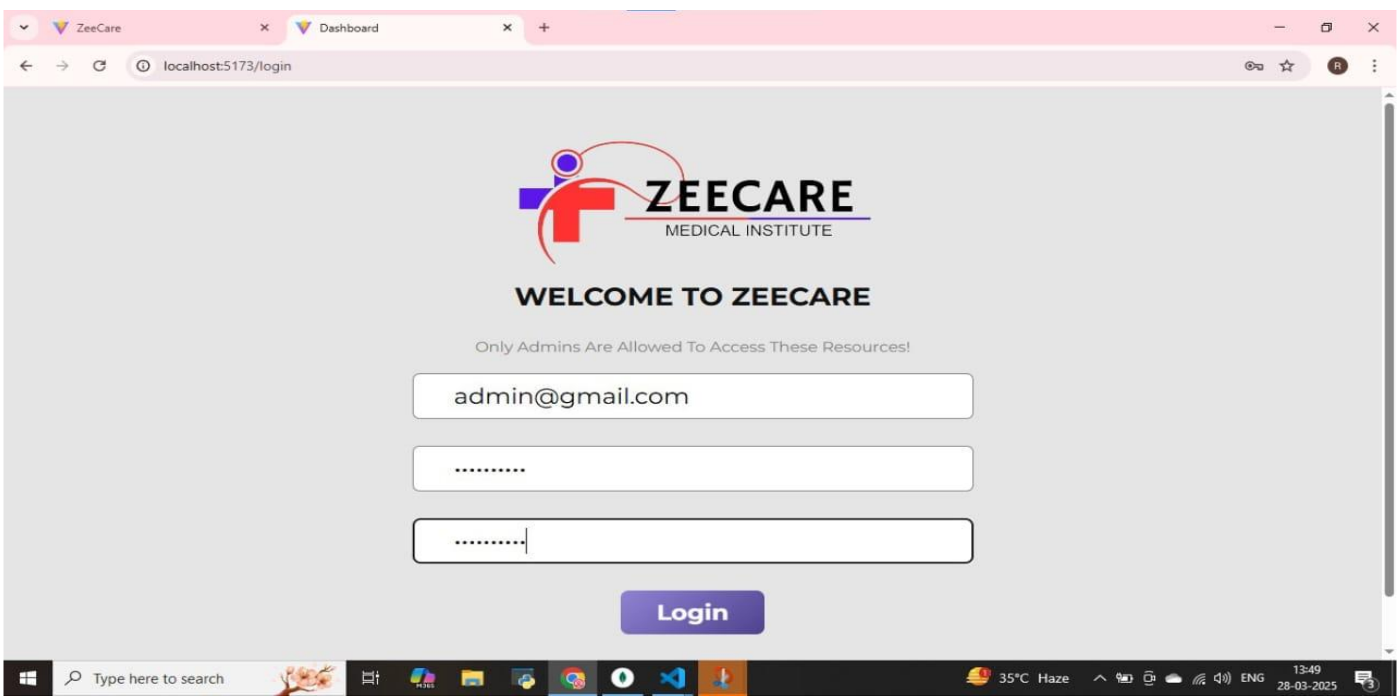
HOME PAGE



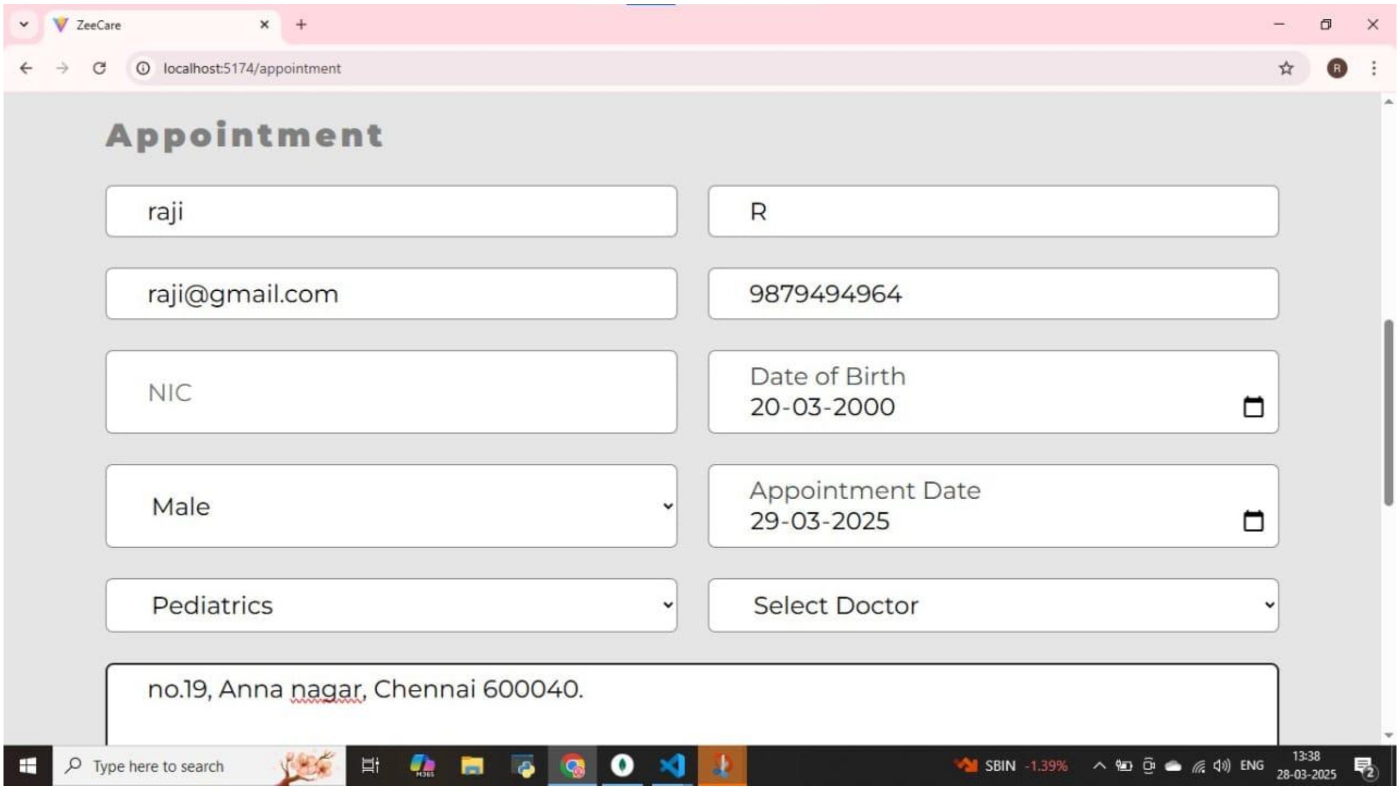
SIGN UP PAGE



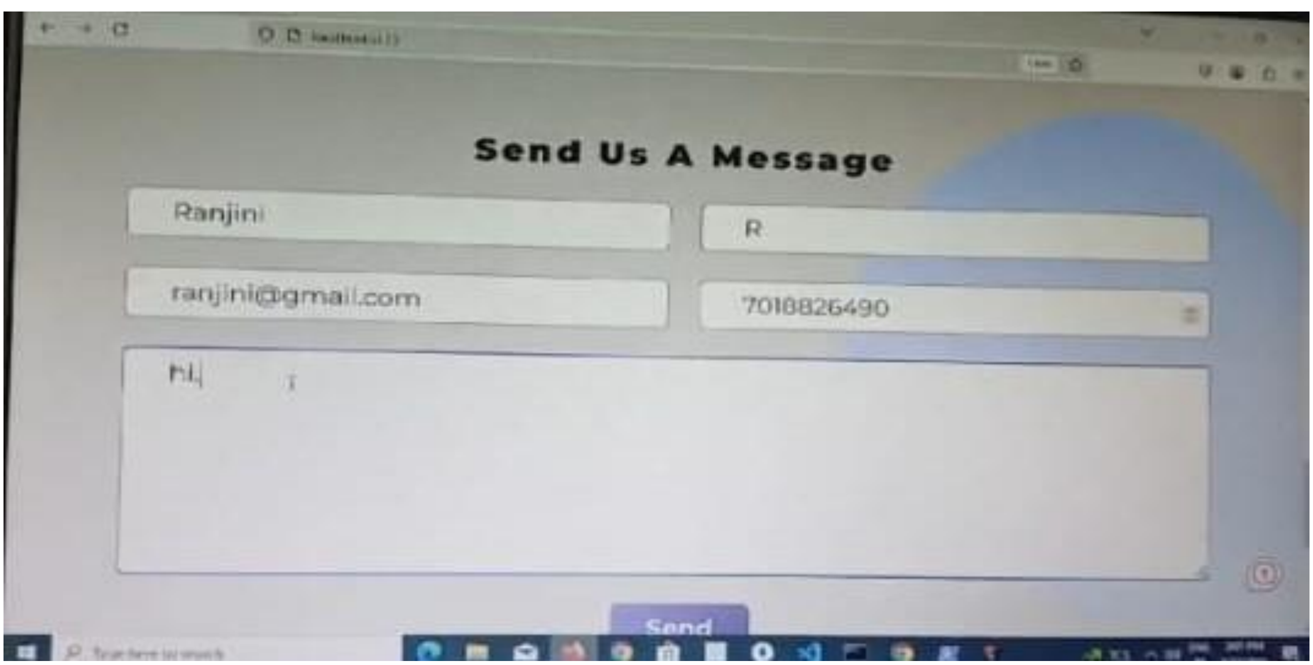
LOGIN PAGE



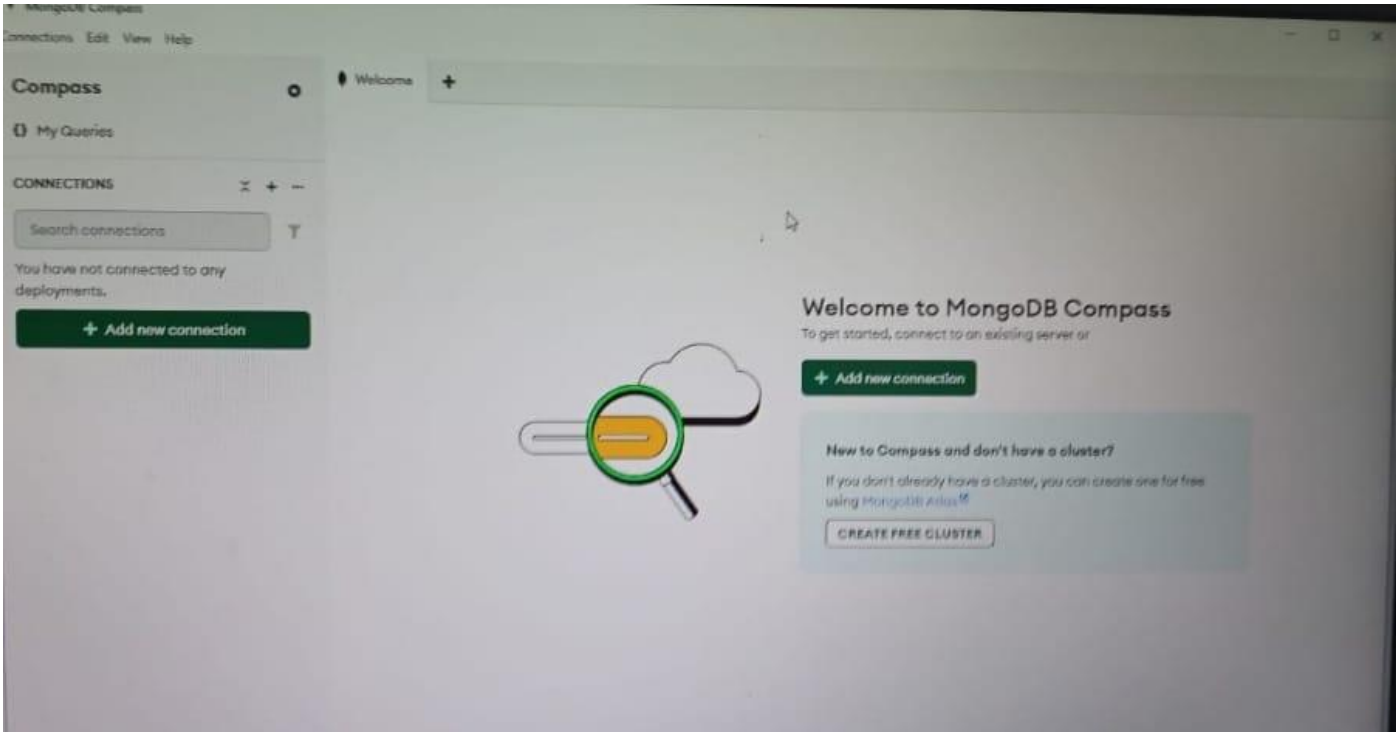
APPOINTMENT PAGE



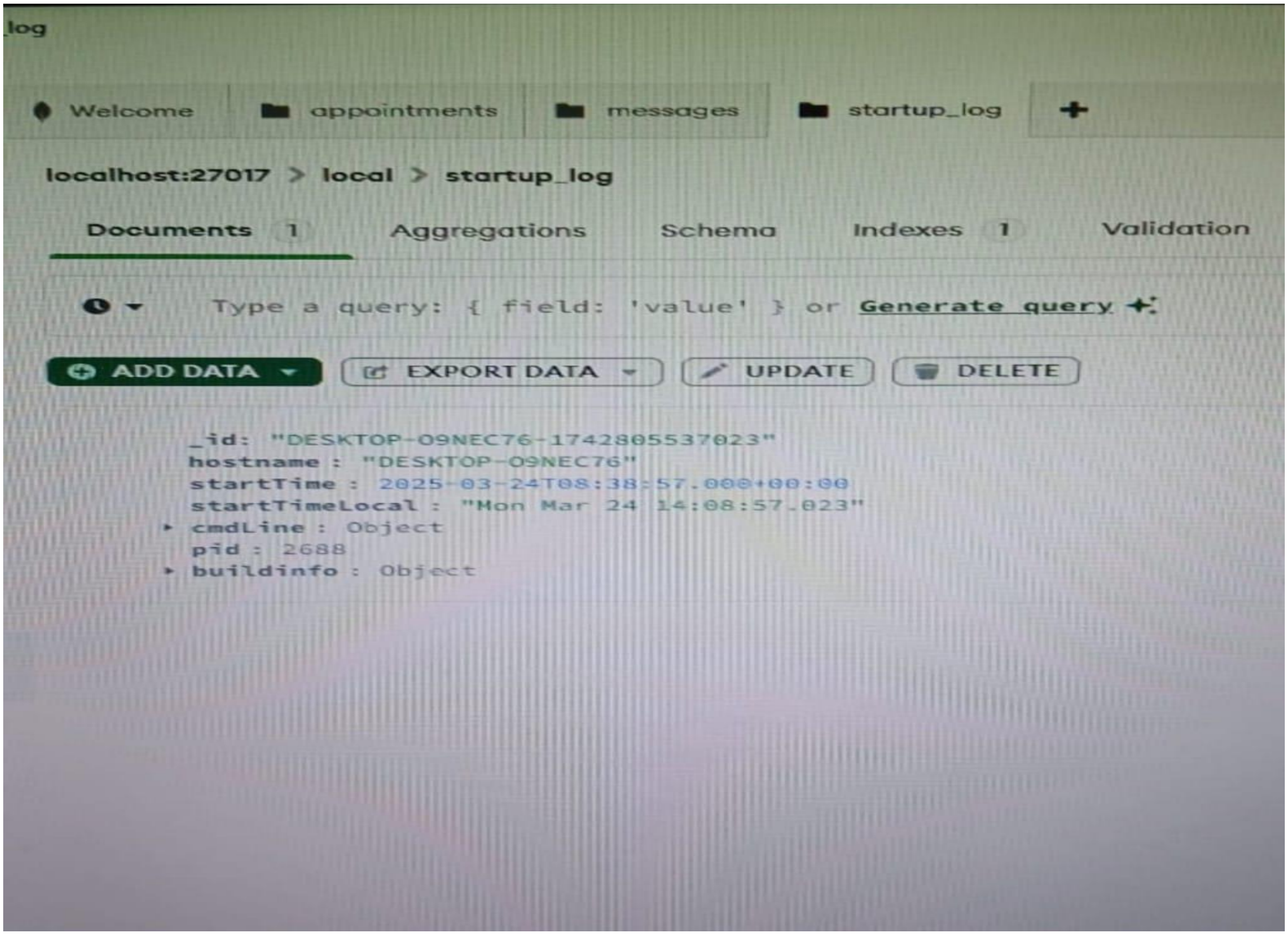
MESSAGE PAGE



MONGODB DATABASE



ADMINISTRATOR PAGE


_log
Welcome
appointments
messages
startup_log
localhost:27017 > local > startup_log
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
EXPORT DATA
UPDATE
DELETE
_id: "DESKTOP-O9NEC76-1742805537023"
hostname : "DESKTOP-O9NEC76"
startTime : 2025-03-24T08:38:57.000+00:00
startTimeLocal : "Mon Mar 24 14:08:57.023"
cmdLine : Object
pid : 2688
buildinfo : Object

TESTING

TESTING

TESTING

Testing is crucial to ensure the system's functionality, security, and performance. Various testing methods are applied to verify the effectiveness of all features and components.

1. Unit Testing

- Tests individual modules like user registration, login, appointment scheduling, and messaging functionality.

- Ensures database operations such as patient record storage and subscription tracking work correctly.

- Validates backend functionalities, including API responses and data handling in MongoDB.

2. Functional Testing

- Ensures each function, such as booking appointments, doctor-patient communication, and subscription management, works as expected.

- Checks error handling mechanisms, such as invalid inputs and failed transactions.

- Validates that users receive confirmation messages after booking appointments.

3. Integration Testing

- Tests interactions between different modules, ensuring seamless connectivity between the frontend (HTML, CSS, JavaScript) and backend (Node.js, MongoDB).

- Ensures real-time updates between patient bookings, doctor availability, and admin controls.

- Verifies that appointment data is correctly stored and retrieved from the database.

4. User Interface (UI) Testing

- Ensures the platform is visually appealing and easy to navigate.
- Tests responsiveness across different devices (mobile, tablet, desktop).
- Checks the proper functioning of buttons, forms, and notifications.

5. Security Testing

- Verifies authentication and authorization mechanisms to prevent unauthorized access.
- Tests encryption methods for securing patient data and communication.
- Identifies vulnerabilities such as SQL injection, cross-site scripting (XSS), and brute-force attacks.

6. Performance Testing

- Measures system response time under different load conditions.
- Evaluates database performance and query execution speed.
- Ensures that multiple users can book appointments simultaneously without system crashes or slowdowns.

7. Usability Testing

- Conducts real-user testing to evaluate ease of use, system efficiency, and accessibility.

- Identifies and resolves UI/UX issues to improve user experience.
- Collects user feedback to refine the system for better engagement.

8. Compatibility Testing

- Ensures the application runs smoothly on different browsers (Chrome, Firefox, Edge, Safari).
- Tests functionality across various operating systems (Windows, macOS, Linux).
- Verifies proper rendering on different screen sizes and resolutions.

9. Regression Testing

- Ensures that new updates or bug fixes do not break existing features.
- Re-runs previous test cases after modifications to confirm system stability.
- Checks backward compatibility with older versions of the application.

10. Database Testing

- Validates data consistency, integrity, and security in MongoDB.
- Ensures correct storage and retrieval of patient details, appointment records, and subscription statuses.
- Checks for database indexing and query optimization for better performance.

FUTURE ENHANCEMENTS

FUTURE ENHANCEMENTS

One major enhancement planned for this system is the integration of AI-powered chatbots to assist patients with basic medical queries before connecting them with a doctor. These chatbots can provide preliminary health advice, suggest common treatments, and even guide users in selecting the right specialist. This feature will reduce doctors' workload by handling minor inquiries and improve response times for urgent cases.

Another key improvement is the addition of video consultation functionality, allowing doctors and patients to interact face-to-face. While text and phone consultations are effective, video calls will enhance communication, enabling doctors to visually assess patients' symptoms. This feature will be particularly useful for conditions requiring physical observation, such as skin disorders or post-surgical follow-ups.

To streamline the prescription process, the system will incorporate an electronic prescription feature, where doctors can generate and send digital prescriptions directly through the platform. Patients will be able to download or share these prescriptions with pharmacies, making the medication procurement process faster and hassle-free.

Enhancing appointment scheduling with AI-based recommendations is another planned feature. The system will analyze patient history and suggest suitable time slots, reducing appointment conflicts and optimizing doctor availability. This enhancement will improve efficiency and ensure better time management for both patients and doctors.

Integration with wearable health devices such as smartwatches and fitness trackers is also under consideration. This will allow doctors to monitor real-time patient data, including heart rate, blood pressure, and oxygen levels. By analyzing this data, doctors can make more informed decisions, especially for patients with chronic conditions like diabetes or hypertension.

Lastly, to expand accessibility, the system will be enhanced with multilingual support, making it easier for non-English speakers to use. Additionally, improved security measures, such as end-to-end encryption and biometric authentication, will be implemented to safeguard patient data and prevent unauthorized access.

APPENDIX

APPENDIX

LIST OF FIGURES:

CONCLUSION

CONCLUSION

The online medical consultation and subscription app provides an efficient and accessible platform for patients to book appointments and consult with doctors through text messages or phone calls. By eliminating the need for physical visits in many cases, the system enhances convenience, especially for individuals with limited mobility or those residing in remote areas. The integration of MongoDB for data management ensures secure and scalable storage of patient records, doctor availability, and appointment details.

This system streamlines communication between patients and healthcare providers, reducing wait times and improving the overall efficiency of medical consultations. Patients can easily schedule, reschedule, or cancel appointments, while doctors can manage their schedules and provide timely responses. The subscription model adds value by offering flexible consultation plans tailored to patient needs, ensuring continuity of care.

Security and data privacy are prioritized in the system, ensuring that sensitive medical information is protected. By implementing authentication mechanisms, only authorized users can access relevant data, reducing the risk of unauthorized access. Additionally, the administrative panel allows efficient management of patient and doctor records, appointment approvals, and system monitoring, maintaining the integrity of the platform.

Looking ahead, future enhancements could include AI-driven chatbots for preliminary consultations, video call support for virtual visits, and integration with electronic health records (EHRs) to provide a more comprehensive healthcare solution. As technology evolves, this platform can continue to improve, making healthcare more accessible, efficient, and user-friendly for both patients and medical professionals.

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REFERENCES

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https://www.ijhep.org/article/15/3/142

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